Project Report for
Chulalongkorn University's Academic Year 2025, Semester 2 offering of
2110215 PROG METH I

JAVA IS DEFEAT

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Introduction

Java Is You is a fan game of Baba Is You, a puzzle game. Both Baba Is You and Java Is You centers around a player character moving tiles in the play area of a level to “manipulate” the behavior of each object in the play area so the player character is able to reach a specified goal point.

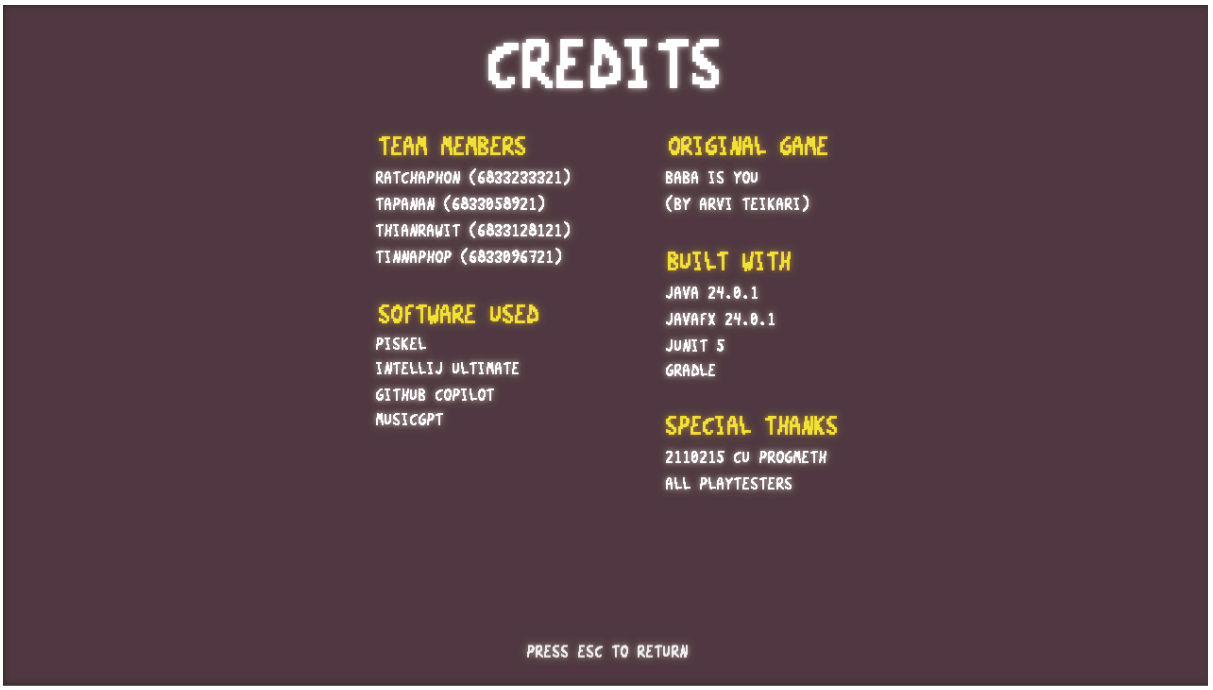
Basic Gameplay

Main Menu

After starting Java Is You, the title screen will be shown.

The player can use the Up or Down keys or the mouse cursor to select an option.

Selecting the Credits option brings the player to the credits screen.



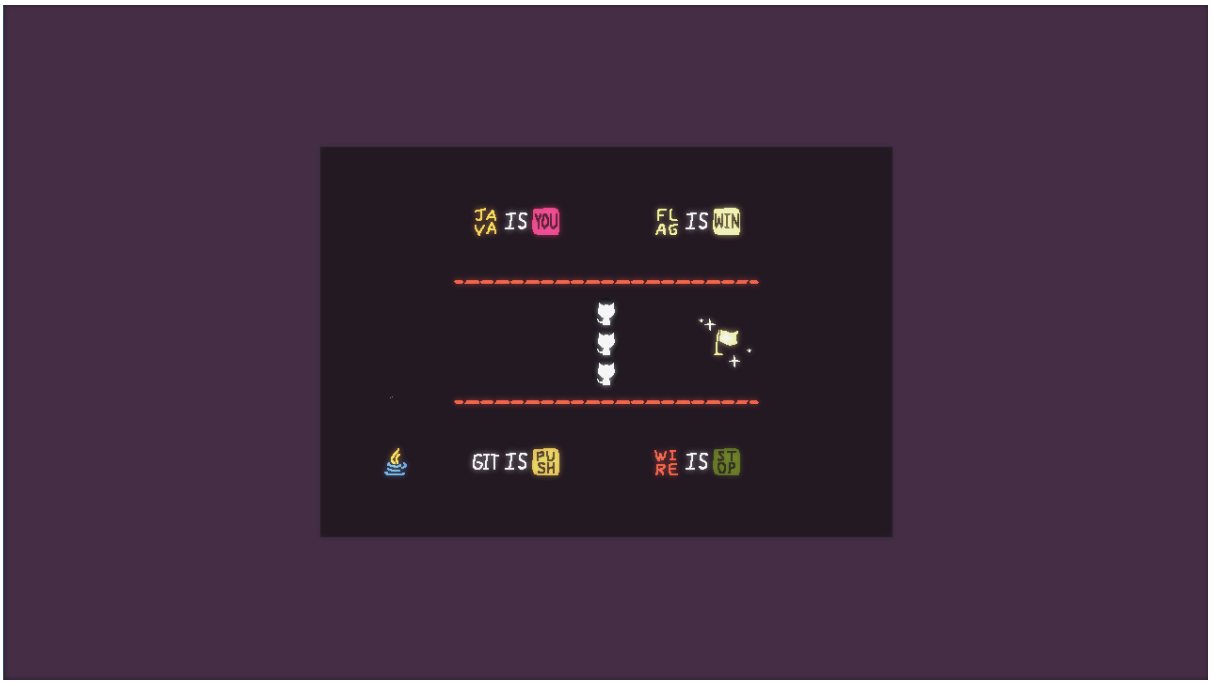
Selecting the Exit the Game option closes the program.
Selecting the Start Game option brings up the level selection menu.

Level Selection Menu

On the level selection menu, the player can use the arrow keys to move the player character (the Java logo) to a tile with a wire on it.
If the player is on a level tile, the player may press Enter to start the level. (Note that levels do not need to be completed in order.)

In a Level

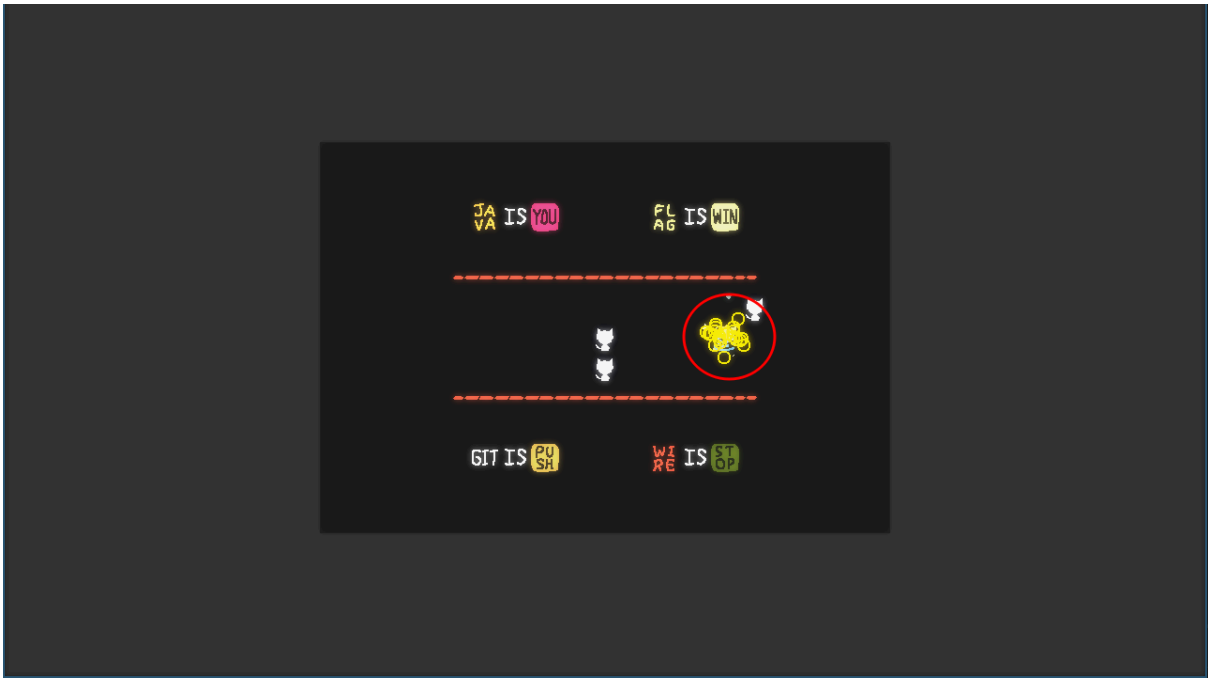
While in a level, the player can use the arrow keys to move their character.



The main goal of the game is to move the player character (using the arrow keys on the keyboard) to its designated finish position (usually a flag or the object designated by the IS WIN sentence).

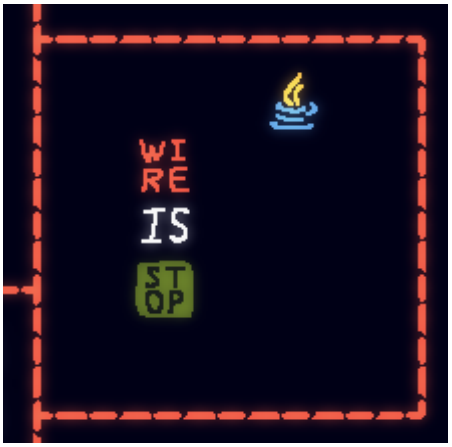
While in a level, the player may also press the Escape key to bring up the pause menu. In the pause menu, the player can resume play, restart play, return to the level selection screen, or return to the main menu. Additionally, on the pause menu, the list of active rules will be displayed.

Upon completion of a level, yellow particles will appear, and the player will be taken back to the level selection screen. Note that level completion does not persist between different playing sessions.



Rules

The rules for Java Is You are similar to Baba Is You.
The player is shown a screen of objects, characters, and pushable word tiles. If three or more word tiles that form a “valid” sentence are aligned together (either formed by the player or is already there initially), they create a rule on how an object behaves. This will be enforced until the word tiles are broken up by (e.g.) pushing the word tiles to break the sentence.



In this picture, WIRE IS STOP.
This indicates that the wires are to stop the player character (the Java logo) from moving across the wire, as its behavior is to stop movement…

…but once that rule is no longer enforced (in this case, WIRE IS is only two word tiles), the player character can move into and across the wire, as it is no longer meant to stop movement.

Word Tiles

Word tiles can be broken into 4 categories: nouns, verbs, properties, and conditions.

Nouns

Noun	Effect
JAVA	Represents an entity. Each word represents a different entity. These entities don’t have any inherent meaning. Only when there is a rule applied to them will they act as a functioning entity.
PYTHON	
XORGATE	
WIRE	
LAVA	
WATER	
FLAG	
DOCUMENT	
TILE	

WARNING	
RECORD	
ERROR	
CHECK	
BUG	
DATABASE	
GIT	
CHIP	

Verbs

Verb	Effect
IS	If a noun and a property surrounds the tiles, the noun gets the attribute represented by the property. If a noun and a noun surround the tiles, all the objects of the type represented by the first noun get replaced by the object type represented by the second noun.
HAS	Whenever the object represented by the first noun is destroyed, create an object represented by the second noun in its place.
EXTEND	Every object represented by the first noun inherits every rule related to the second noun

Properties

Property	Effect
YOU	Designates the object type represented by the noun of the sentence to be the player character.
WIN	Designates the object type represented by the noun of the sentence to be the winning location.
DEFEAT	Designates the object type represented by the noun of the sentence to kill all the player in the same tile.
PUSH	Designates the object type represented by the noun of the sentence to be pushable.
STOP	Designates the object type represented by the noun of the sentence to stop movement of other objects.
SINK	Gives the object type represented by the noun of the sentence that is on the ability to sink other objects.

	When sinking an object, destroy both the object on the tile, and the tile that sunk the object.
HOT	Designates the object type represented by the noun to be able to melt an object that is MELT. When melting an object, destroy the object on the tile.
MELT	Designates the object type represented by the noun of the sentence to be destroyed upon being in the same position as a HOT object.
MOVE	Designates the object type represented by the noun of the sentence to be able to move (along a predetermined path). The object will move one step per one player movement/keypress.
MORE	Designates the object type represented by the noun of the sentence to create a copy of itself in 4 directions each turn.

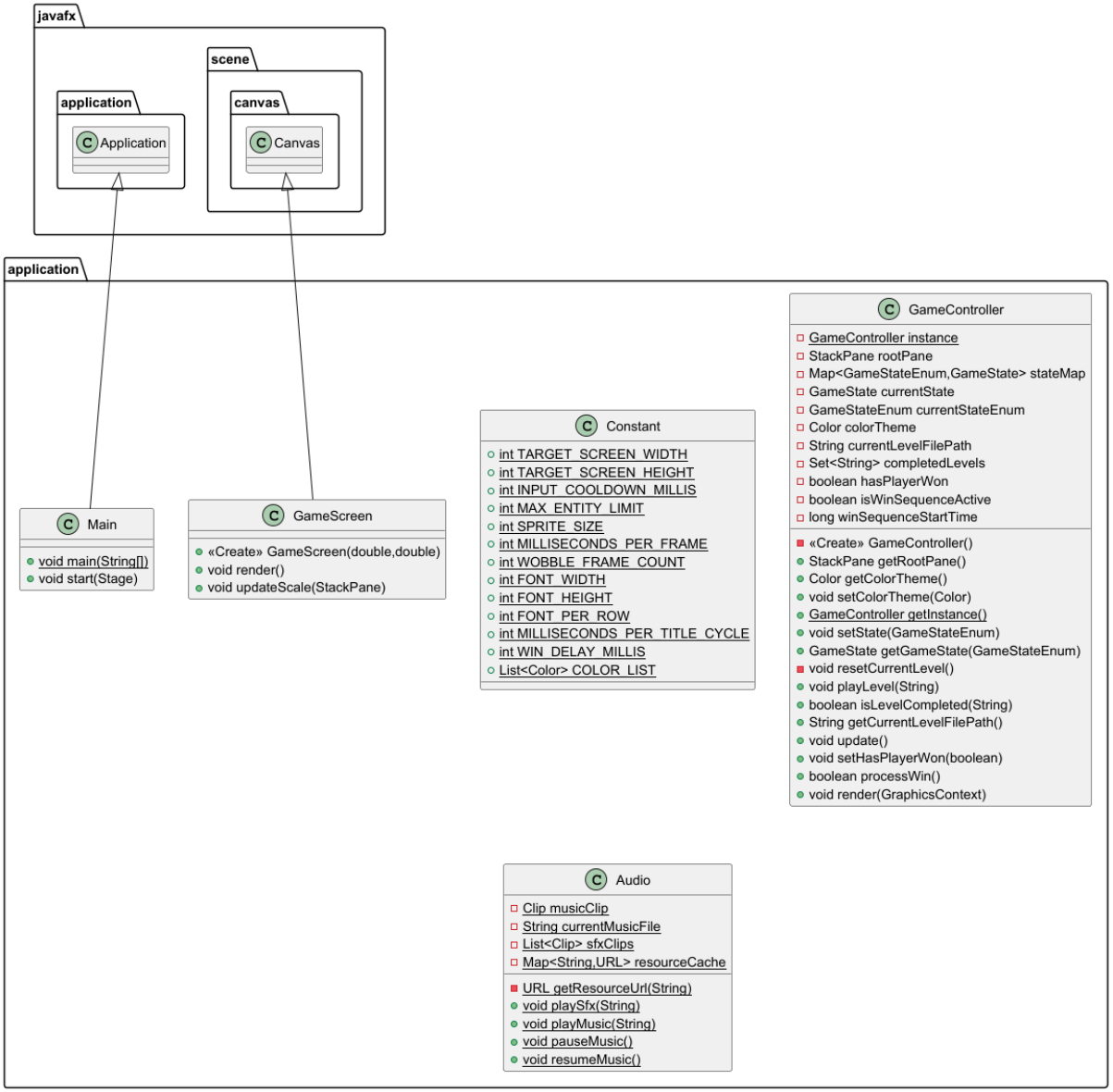
Conditions

Condition	Effect
ON	Only activates a sentence if an/the object type represented by the first noun on the same position of the a/the object represented by the second noun.
NEAR	Only activates a sentence if an/the object type represented by the first noun is in the 8 tiles around the position of the a/the object represented by the second noun.
FACING	Only activates a sentence if an/the object type represented by the first noun is facing a/the object represented by the second noun.
INSTANCEOF	Only activates a sentence if an/the object type represented by the first noun is an instance of the second noun in the inheritance chain created by A EXTEND B rules.

UML Diagrams

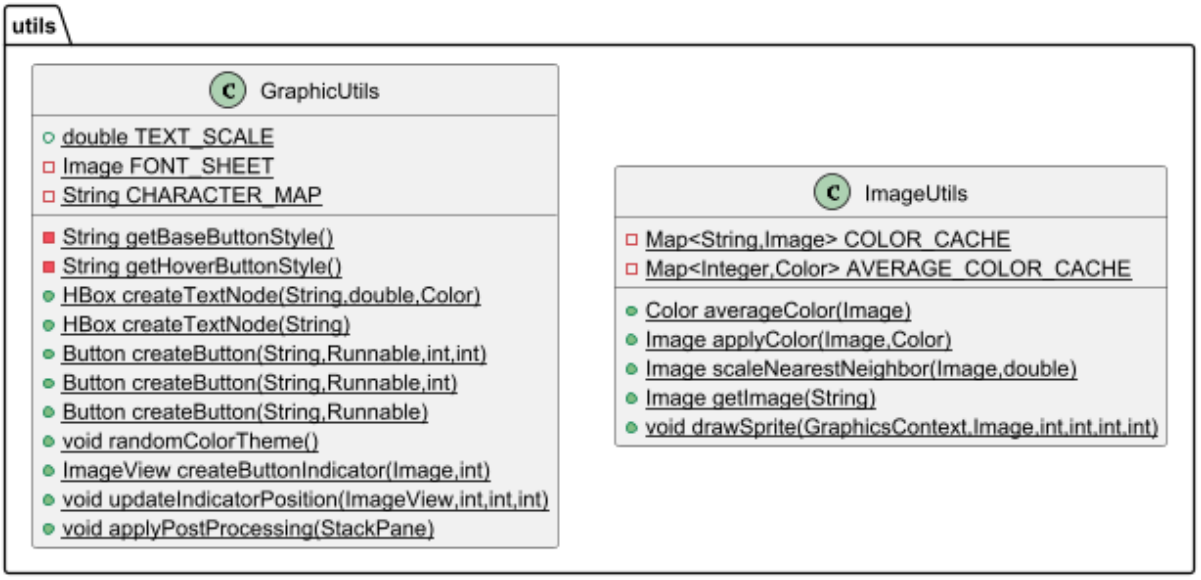
Package Application

The package Application is the core of the game, responsible for storing universal constants required for the game's functionality, setting up the game screen and game logic, and starting the game itself.



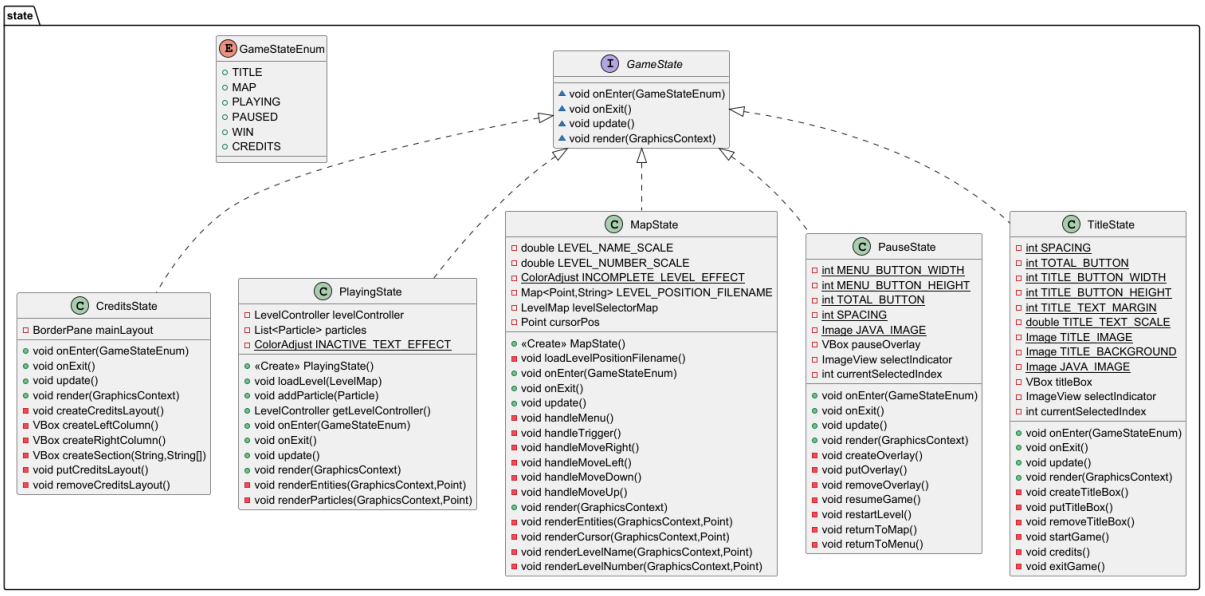
Package Utils

The package Utils contains miscellaneous UI and rendering helpers such as image loading, button creation, text creation, post-processing, image coloring and scaling, etc.



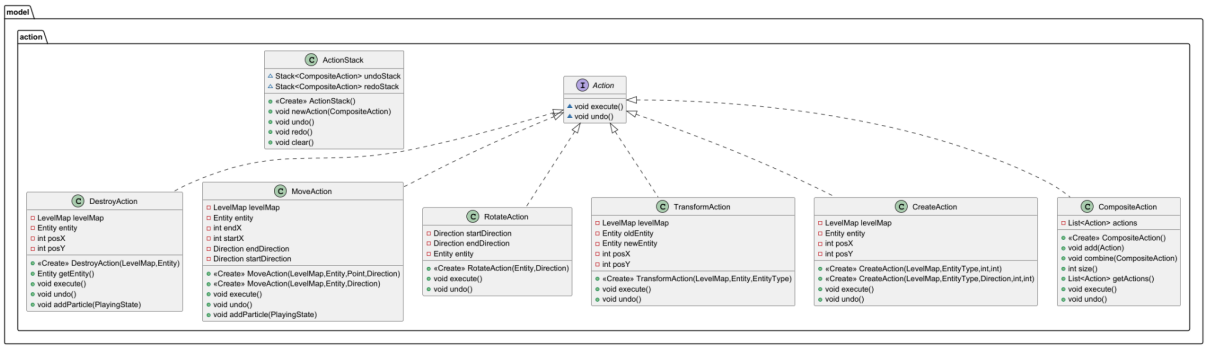
Package State

The package State implements the game states and UI screens, state transitions, and rendering life-cycles. It takes player input received from the logic.input package and use it to control the game.



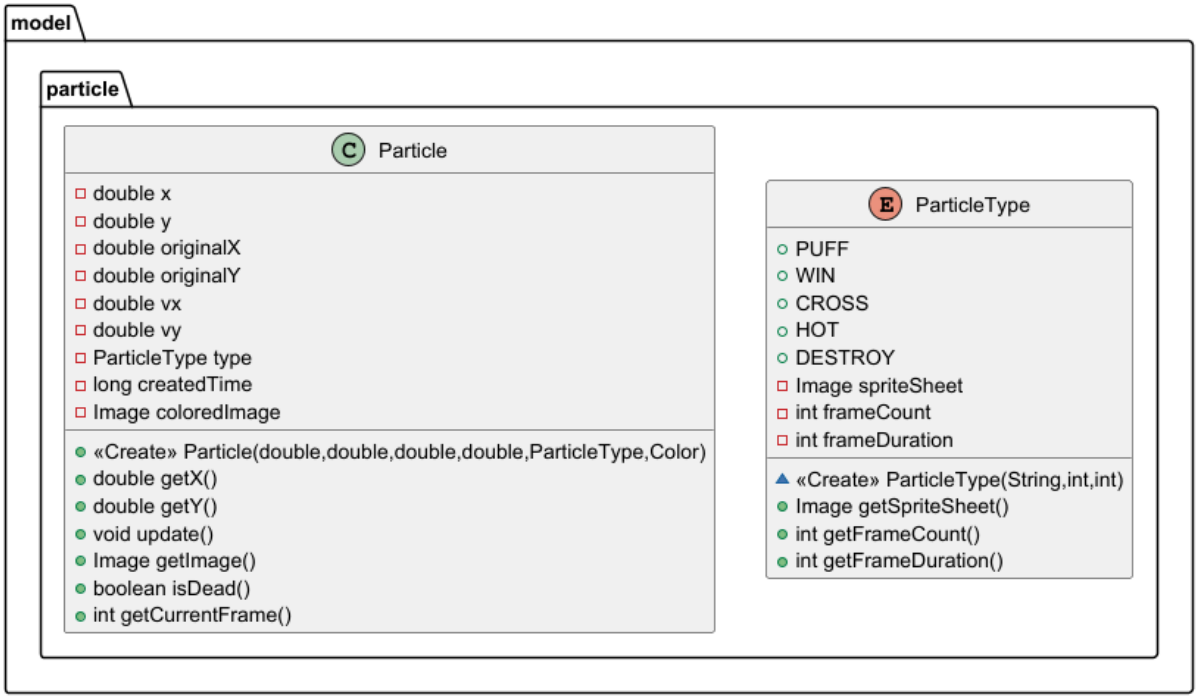
Package Model.Action

The package Action contains the Action command pattern for game undo/redo. An action can be executed to modify the LevelMap and then stored, so it can be used to undo/redo later.



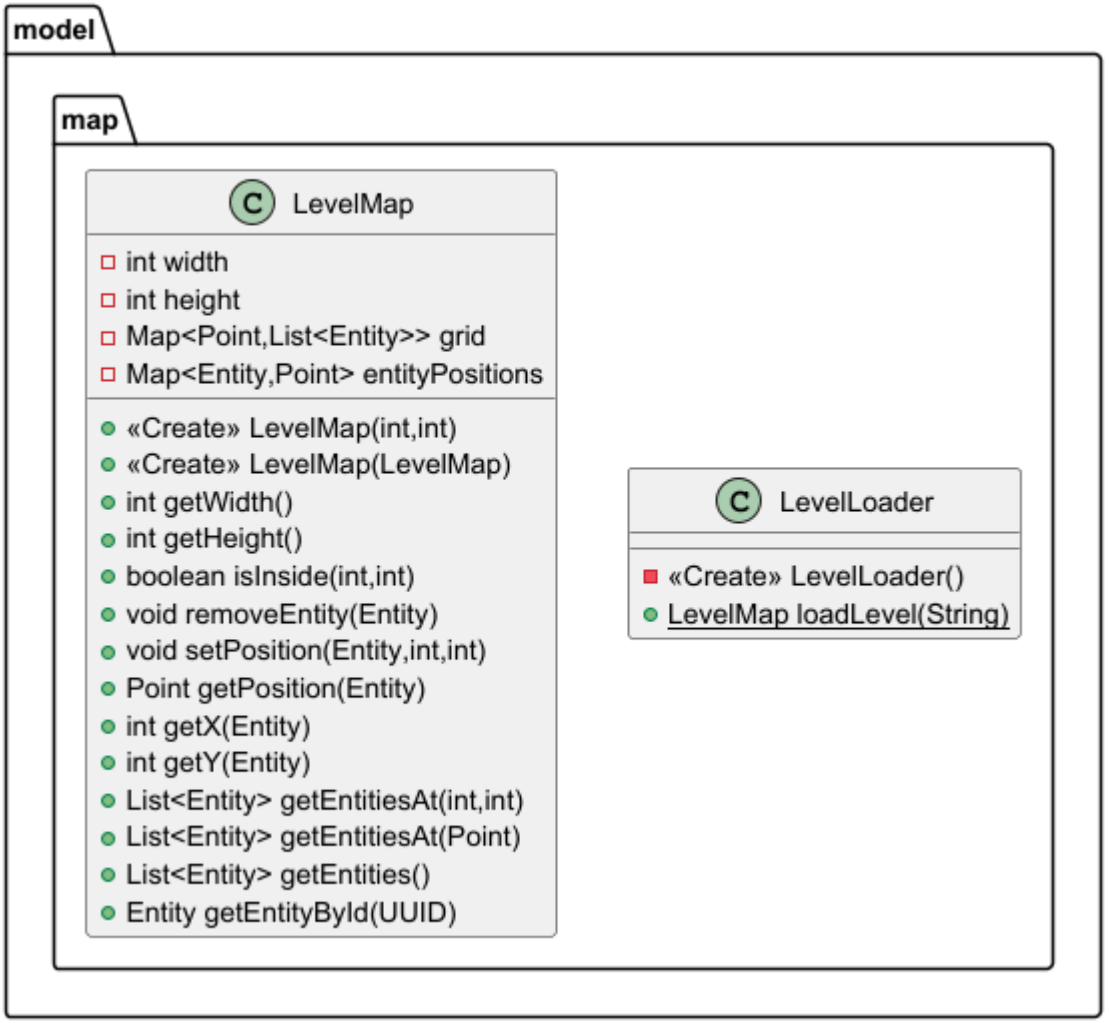
Package Model.Particle

The package Particle contains Particle types, runtime particle objects, and their update logic for visual effects during gameplay.



Package Model.Map

The package Map contains the data structure of level data with public methods for manipulating it, and the LevelLoader which reads and translates the .csv file text into in-game entities.

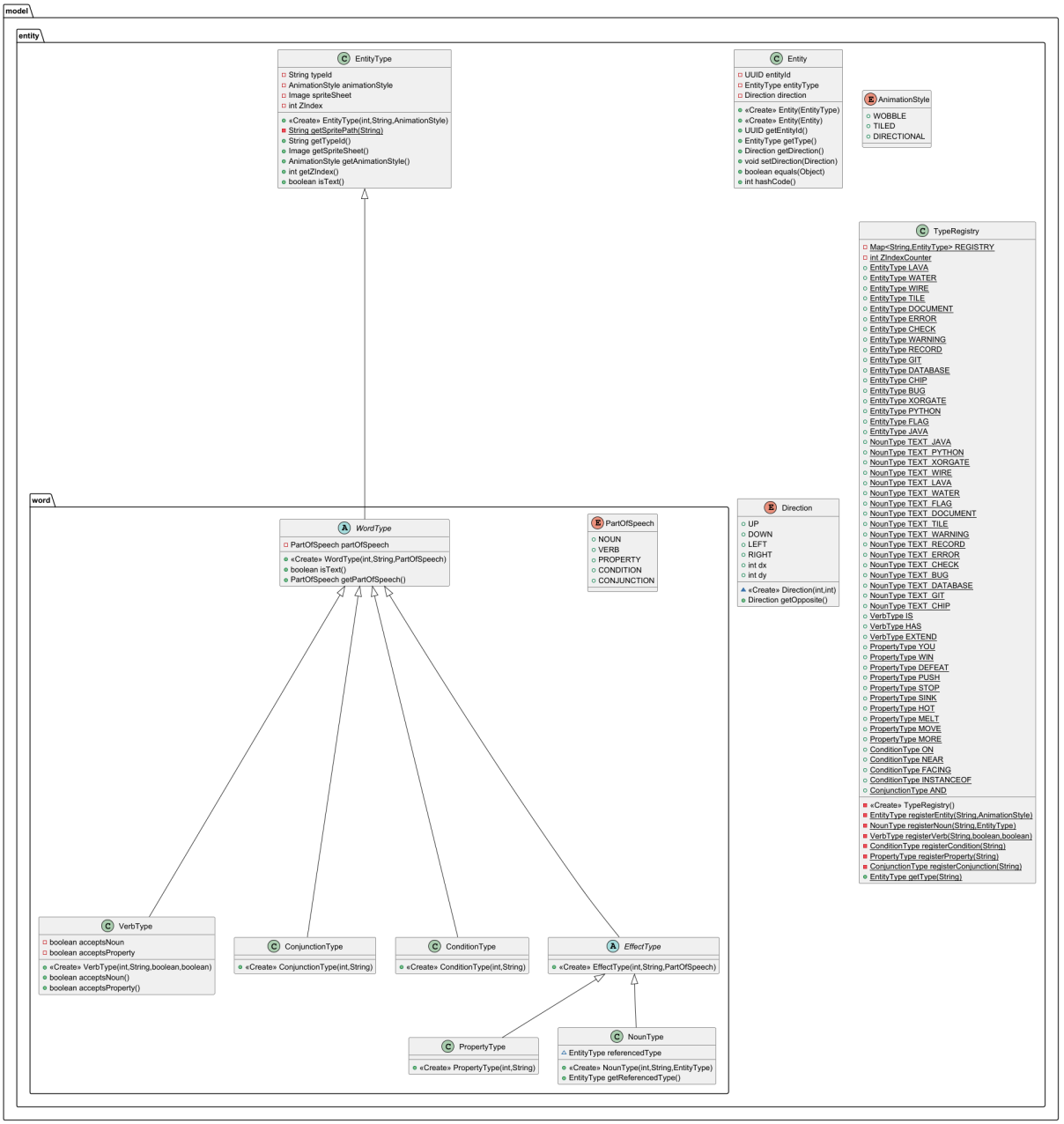


Package Model.Rule

The package Rule contains the rule model object and other rule-related models. Since the rules can be created or destroyed in-game, a dynamic representation of any rules is needed, and implemented in the form of subjct, verb, effect, and conditions.

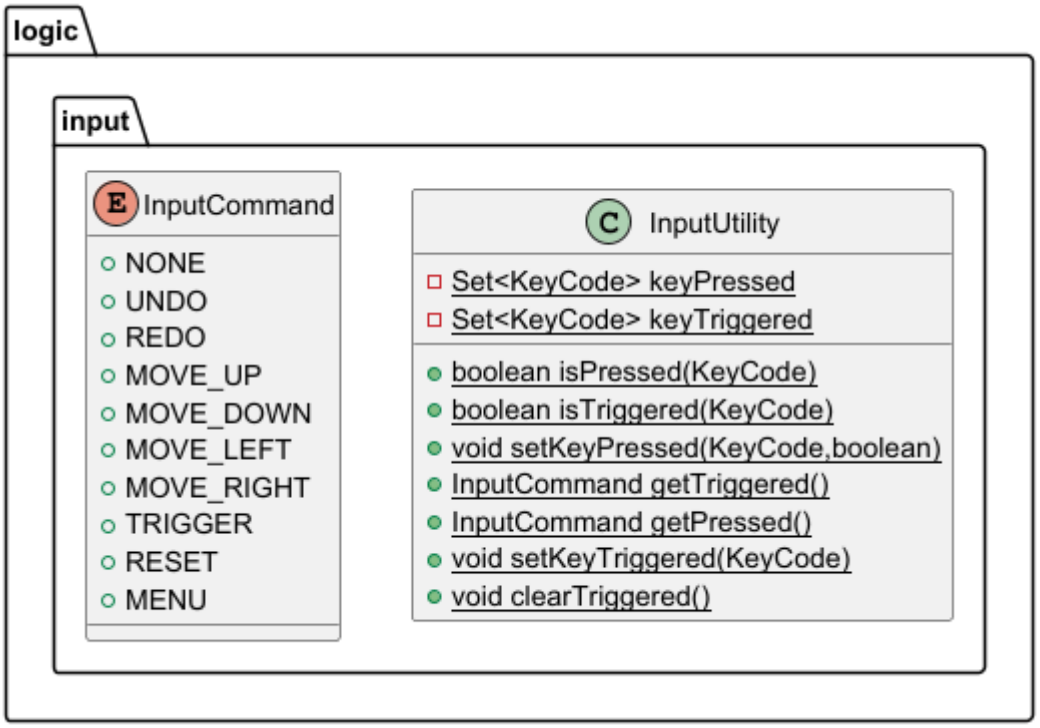
Package Model.Entity

The package Entity contains a core entity/type system for game objects. Every entityType is inherently the same, except word types that can form rules, so this package uses type object pattern and inheritance for words to store additional information for each word type. The TypeRegistry is the central class to register all available entity types.



Package Logic.Input

The package Entity contains the logic for input mapping and utilities for transforming keyboard events into game-level input commands.

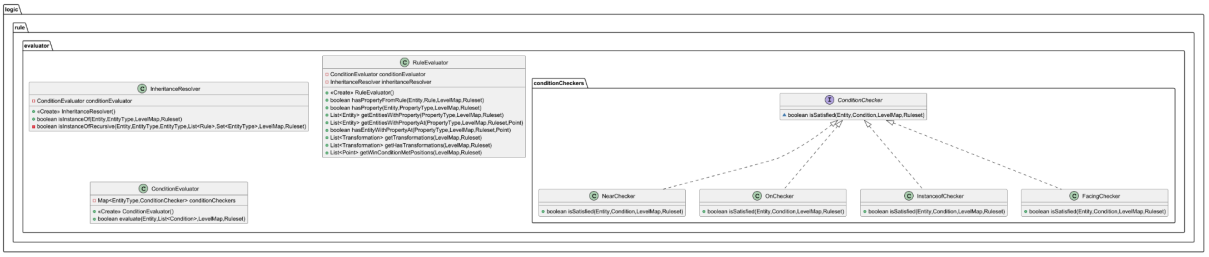


Package Logic.Level

The package Level includes high-level level control and turn orchestration. Running a turn requires a multiple step pipeline to get the correct next state according to the rule and input.

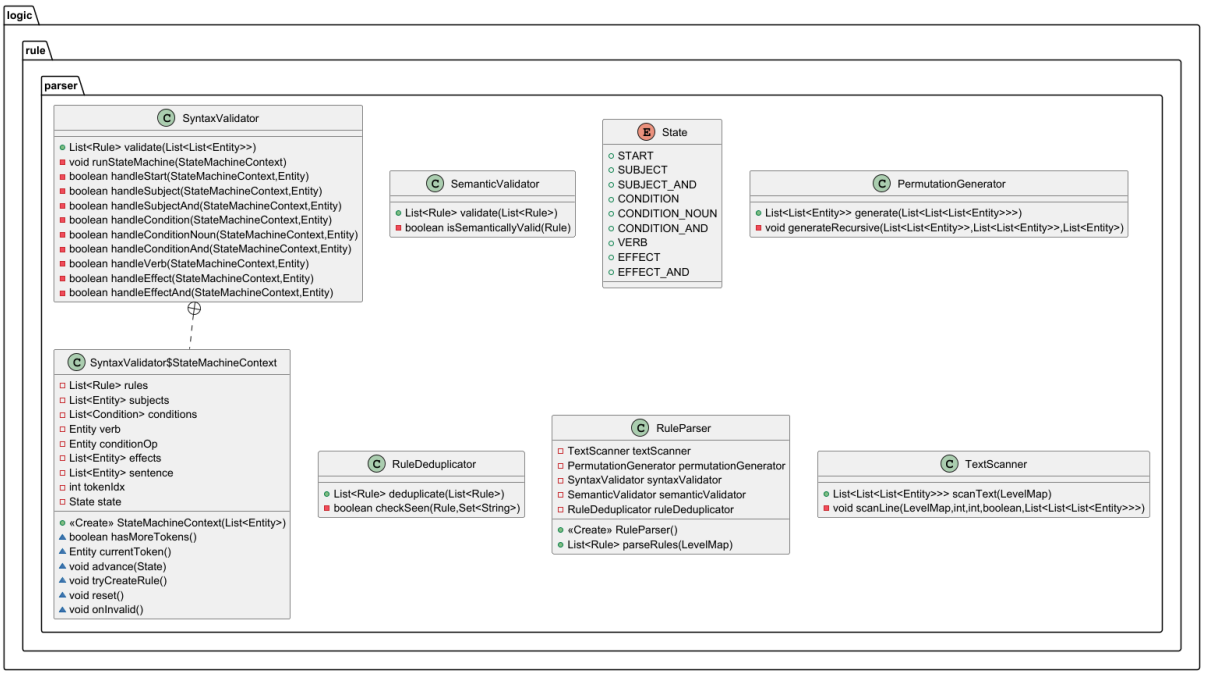
Package Logic.Rule.Evaluator

The package Evaluator is the rule and effect evaluation engine which evaluates the rule against the LevelMap to query properties, compute transformations, and resolve inheritance.



Package Logic.Rule.Parser

The package Parser contains the rule parsing pipeline to create current rules from LevelMap state. It starts with text scanning to find potential rules, then generates all permutations for stacked texts, then runs the rule through a state machine to filter out invalid syntax, and then finally removes duplicate and semantically invalid rules.



JavaDoc

The methods of each class and its purpose can be found in the documentation, which is at https://storage.mistertfy64.com/java-is-you/documentation/index.html

Bibliography

https://en.wikipedia.org/wiki/Baba_Is_You (accessed March 2, 2026)
https://babaiswiki.fandom.com/wiki/Category:Nouns (accessed March 2, 2026)
https://babaiswiki.fandom.com/wiki/Category:Properties (accessed March 2, 2026)